

Agenda and Minutes

Chemical Engineering Curriculum Day – May 17, 2022

EIB Room 224 (Conference Room) and remote access

Agenda:

1. Vikas and Alan: Brief comments on curriculum day
2. Alan Zdunek: Introduction to the current curriculum and all its components
3. Betul Bilgin: Survey results from senior students
4. Alan Zdunek: Discussion with faculty on the syllabus of the 200 level classes.
5. Lunch
6. Alan Zdunek: Discussion with faculty on the syllabus of all the 300 level classes
7. Alan Zdunek: Discussion with faculty on all the 400 level classes
8. Vikas Berry: Wrap up and setting up a summer committee on "Introduction to Chemical Engineering"

Notes:

Agenda items in red were discussed in the meeting. For the 300-level courses, 301, 311, 312 were discussed. Two, 400-level courses, CHE 451 and CHE 499 were also discussed as well as ENGR101. The rest of the 300-level courses and 400-level course will be discussed at another Curriculum Day meeting.

Introduction to Current Curriculum (Zdunek):

Presented the BS in ChE degree program: 128 credit hours covering: non-engineering and general education requirements, required engineering courses; ChE and non-ChE, technical electives, and concentrations to the BS ChE degree

Gave an overview of ENGR101 (1st-year and incoming transfer student orientation course) and CHE 499 Professional Development seminar.

Faculty suggested emphasizing importance of math skills in chemical engineering education.

Senior Exit Survey results (Bilgin):

Betul will send a report on the results.

Some student comments:

CHE students need more space to study in EIB. Student lounge okay but can be noisy, library very small.

Vikas suggested 2nd floor room next to EIB 271 might be available in the future.

CHEM 342 is not very useful since ChE students take CHE 301

CHEM 234 not very useful and topics are pre-med oriented

CHEM 222 needs to be improved. Equipment out-of-date, similar topics as general chemistry

Math and Physics courses are okay

Zdunek to look into ABET requirements on how many math/science courses are needed in BS ChE major.

Students were struggling with CHE 311 and 312. They need more preparation.

Students suggested 433 as a permanent course.

CHE courses:

200-Level Courses

CHE 201 Introduction to Engineering Thermodynamics (Instructor: Betul Bilgin)

Discussed syllabus

Good syllabus template with topics such as Academic Integrity, Diversity, Equity, Inclusion, College Resources, etc. Will provide for ChE faculty to use.

Currently using Wiley 'Fundamentals of Engineering Thermodynamics' (has been used for many years)

Switching to 'Engineering and Chemical Thermodynamics' Koretsky and Liberatore next Fall

Uses [zybook \(https://www.zybooks.com/catalog/?fwp_subjects=chemical-engineering\)](https://www.zybooks.com/catalog/?fwp_subjects=chemical-engineering), Gradescope and Piazza discussion format

Projects/activities- 4 to 5 per semester

AICHE safety module assignment (e-learning)

Homework, Quizzes, Exams, Final Exam, activities/projects

CHE 205 Computational Methods in Chemical Engineering (Instructors: Shafigh Mehraeen, Betul Bilgin)

Discussed syllabus: Good agreement between topics covered by Mehraeen and Bilgin

Some of the topics covered: Linear algebra, non-linear equations, Solving Differential and Partial differential equations

Discussion that students need more background in math to be able to understand the computational methods of solving DE's and PDE's. Suggestion is to make MATH 210 or MATH 220 a co-requisite (concurrent). But might cause a bottleneck in course flow for students.

Focus is on Excel/VBA as a computational method, but not too much MatLab

Discussion and strong agreement that a computing course before CHE 205 is needed, CS109 should be a pre-requisite .

Discussion about current topics covered in CS109- Python instead of C++ and MatLab.

Zdunek to check with EPC about what CS will teach in CS 109.

If the students learn Python, will this give them sufficient knowledge for CHE 205?

Faculty agreed that learning one language is sufficient.

Comments that Python is being used more frequently in industry. Another comment that Excel/VBA still used quite a bit in industry.

Homework, quizzes, exams, final exams, projects

CHE 210 Material and Energy Balances (Instructor: Lew Wedgewood)

Discussed syllabus and topics

Introduction to CHE processes and economics

Material balances, reactive processes in multi-component systems

Energy balances in same

Simple shell balances covered

Textbook used: Introduction to Chemical Processes: Principles, Analysis, Synthesis by Regina M Murphy

Most up-to-date book on the subject compared to other traditionally used books.

Instructor assumes student know how to use Excel, appears to be okay.

Industry professionals come to class and suggest industry-relevant problems (part of NSF Education grant)

Discussion:

1. Taking 201 and 210 concurrently, or should 201 be a hard pre-requisite for 210?
2. Taking 210 and 311 concurrently, or should 201 be a hard pre-requisite for 311?

Ideally take 201 → 210 → 311

Understanding of a shell balance is the main issue for 210 to be a pre-requisite to 311 and was recommended.

Homework, Quizzes, Exams, Final Exams

CHE 230 Molecular Systems in Chemical Engineering (Instructor: Ezinne Achinivu)

Discussed syllabus and overall concept of course: this is the first semester course has been offered.

Instructor emphasized three overall concepts that branched out into specific topics: molecular design, structural properties, applications.

Covered biomolecules but needed to provide biology background as students were lacking in this knowledge. Instructor suggested the need for a better biology background.

Used methods to encourage students to read before class and participate in class (Kahoots)

Excel proficiency for numeral computation was expected.

Technical writing proficiency was expected but was weak.

Current pre-requisite is CHEM 232. Instructor suggests adding CHEM 233

Discussion: Molecular engineering is a new discipline and contains a broad area of topics. Due to the broad topic area and newness of the subject, there is not a textbook that can be used to cover all topics. Further discussion needed on defining the key concepts to be covered in the course. How do we tailor this course to cover the key concepts in molecular engineering as an introductory 200-level engineering course? Kind of like how a general chemistry course covers a broad variety of chemistry topics.

How much material property topics should be covered? Will this course substitute for CME 260? Could we switch our BS CHE requirements to include a BME course instead of CME 260?

CHE 301 Chemical Engineering Thermodynamics (Instructor: Michael Caracotsios)

Discussed syllabus and Chemical Engineering Calculator

Current Prerequisites: CHE 201, 205

Michael suggested that CHE 210 also be a pre-requisite

Textbook: Michael Caracotsios notes and chapters from several well-known textbooks: Smith & VanNess, Matsoukas, Fundamentals of Chemical Engineering Thermodynamics, Matsoukas

Topics: Chemical engineering thermodynamics principles covered. Does not cover chemical reaction equilibrium, but covers some Special Topics

Assignments: Emphasis on homework problems (instructor generated, not from textbook) for learning and grading

Chemical Engineering Calculator:

Developed over several years with help of undergraduate students,

Uses Excel/VBA and different numerical solvers and methods, has a nice user interface.

Used to solve a variety of chemical engineering problems and information (property calculations, VLE, thermodynamics, transport phenomena, process control, cost estimation).

Not a 'black box' but requires knowledge of the underlying principles behind the calculations.

Very powerful tool, discussed how this may be incorporated into other courses.

Discussion on Intellectual Property regarding the calculator and patenting it.

CHE 311 Transport Phenomena I: Momentum Transport

(Vivek Sharma, Michael Caracotsios, Ying Liu Instructors)

Two different syllabus (Sharma and Caracotsios) presented and discussed

Prerequisites: CHE 205, credit or concurrent registration in CHE 210 and MATH 220

Discussion on making CHE 210 and MATH 220 a hard pre-requisite, instead of concurrent registration. Due to the topics covered in this course, students seem to not have a good understanding of shell balances that they would get from CHE 210. Also differential and partial differential equations knowledge is needed and seems to be lacking in students.

Textbook: Sharma and Liu use Bird, Stewart and Lightfoot, Caracotsios uses Momentum Transport Notes Bird, Stewart, Lightfoot, and Geankoplis "Transport Processes and Unit Operations".

Topics covered: Macroscopic balances, shell balances, Equations of Change, e.g., Navier-Stokes equation, 2D flow, turbulent flow, friction factors. Potential flow and boundary layer flow not covered. Non-newtonian fluids covered by Caracotsios
Sharma and Liu give a mathematics review.

Sharma has 3 grading options, students can skip final if appropriate.

Caracotsios emphasizes homework assignments (instructor generated, not textbook problems)

CHE 312 Transport Phenomena II: Heat and Mass Transport

(Instructor: Lew Wedgewood)

Discussion of syllabus and topics covered.

Textbook: Bird, Steward and Lightfoot, reference textbook: Fundamentals of Heat and Mass Transfer, Incropera and De Witt

Grading: HW assignments, quizzes, 2 exams, final exam

Topics covered:

Math review, history of chemical industry, transport properties overview, Shell balances (heat conduction, mass balance, diffusion with chemical reactions, heat conduction through walls, forced convection, free convection, falling film diffusion, equations of change including dimensional analysis, 2d flow, unsteady flow, turbulence, interphase transport in non-isothermal systems and multi-phase systems, macroscopic energy balances, applications like a packed bed absorber.

CHE 451 Renewable Energy Technologies

(Instructor: Said Al-Hallaj)

Discussed syllabus and topics covered

This is a 400-level technical elective for undergraduates and suitable for graduate students

Grading: weekly assignments, midterm exam, final project and oral presentation

Textbook: Hybrid Hydrogen Systems for Stationary and Transportation Applications, Al-Hallaj and Kiszynski

Topics: Renewable energy sources, energy storage and conversion, hydrogen economy, case studies (Hybrid wind/solar hydrogen, hybrid desalination, fuel cell and solar desalination)